

Ex.no:2	EDA-DATA IMPORT AND EXPORT

Aim:

To inspect dataset features, detect null values, and export data into CSV, Excel, and SQLite database formats.

Algorithm:

1. Import pandas and sqlite3 libraries.
2. Read hotel_bookings.csv using pd.read_csv().
3. Inspect data structure using .head(), .tail(), .info(), .describe(), and .isnull().sum().
4. Export DataFrame to .csv and .xlsx formats using .to_csv() and .to_excel().
5. Connect to hotel_booking.db using sqlite3 and write data into an SQL table via .to_sql().
6. Close the database connection and end execution.

Program:

```
# Import required libraries

import pandas as pd

# Load the Hotel Booking dataset

df = pd.read_csv("hotel_bookings.csv")

# Display the complete dataset

print("===== DATASET =====")

print(df)

# Display first five records

print("\n===== FIRST FIVE ROWS =====")

print(df.head())

# Display last five records

print("\n===== LAST FIVE ROWS =====")

print(df.tail())

# Display dataset information

print("\n===== DATASET INFORMATION =====")

print(df.info())

# Display statistical summary

print("\n===== STATISTICAL SUMMARY =====")

print(df.describe(include='all'))

# Display dataset shape

print("\nDataset Shape:", df.shape)

# Display column names

print("\nColumns:")
```

```
print(df.columns)

# Check missing values
print("\n===== MISSING VALUES =====")
print(df.isnull().sum())

# Export dataset to CSV
df.to_csv("hotel_bookings_export.csv", index=False)
print("CSV file exported successfully!")

# Export dataset to Excel
df.to_excel("hotel_bookings_export.xlsx", index=False)
print("Excel file exported successfully!")

import sqlite3

# Create SQLite database
conn = sqlite3.connect("hotel_booking.db")

# Export DataFrame to SQL table
df.to_sql(
    "hotel_bookings",
    conn,
    if_exists="replace",
    index=False
)
print("Dataset exported successfully to SQLite database!")
conn.close()
```

Output:

===== DATASET =====

	hotel	is_canceled	lead_time	arrival_date_year	\
0	Resort Hotel	0	342	2015	
1	Resort Hotel	0	737	2015	
2	Resort Hotel	0	7	2015	
3	Resort Hotel	0	13	2015	
4	Resort Hotel	0	14	2015	
...	...	...	...	...	
97256	City Hotel	0	23	2017	
97257	City Hotel	0	102	2017	
97258	City Hotel	0	34	2017	
97259	City Hotel	0	109	2017	
97260	City Hotel	0	205	2017	

	arrival_date_month	arrival_date_week_number	arrival_date_day_of_month	\
0	July	27	1	
1	July	27	1	
2	July	27	1	
3	July	27	1	
4	July	27	1	
...	...	...	...	
97256	August	35	30	
97257	August	35	31	
97258	August	35	31	
97259	August	35	31	
97260	August	35	29	

	stays_in_weekend_nights	stays_in_week_nights	adults	...	deposit_type	\
0	0	0	2.0	...	No Deposit	
1	0	0	2.0	...	No Deposit	
2	0	1	1.0	...	No Deposit	
3	0	1	1.0	...	No Deposit	
4	0	2	2.0	...	No Deposit	
...	...	...	...	...	...	
97256	2	5	2.0	...	No Deposit	
97257	2	5	3.0	...	No Deposit	
97258	2	5	2.0	...	No Deposit	
97259	2	5	2.0	...	No Deposit	
97260	2	7	2.0	...	No Deposit	

===== STATISTICAL SUMMARY =====

	hotel	is_canceled	lead_time	arrival_date_year	\
count	97261	97261.000000	97261.000000	97261.000000	
unique	2	NaN	NaN	NaN	
top	City Hotel	NaN	NaN	NaN	
freq	67998	NaN	NaN	NaN	
mean	NaN	0.392274	109.489950	2016.170880	
std	NaN	0.488260	110.676344	0.680581	
min	NaN	0.000000	0.000000	2015.000000	
25%	NaN	0.000000	19.000000	2016.000000	
50%	NaN	0.000000	73.000000	2016.000000	
75%	NaN	1.000000	170.000000	2017.000000	
max	NaN	1.000000	737.000000	2017.000000	

	arrival_date_month	arrival_date_week_number	\
count	97261	97261.000000	
unique	13	NaN	
top	August	NaN	
freq	11244	NaN	
mean	NaN	27.435344	
std	NaN	13.359844	
min	NaN	0.000000	
25%	NaN	17.000000	
50%	NaN	28.000000	
75%	NaN	38.000000	
max	NaN	53.000000	

	arrival_date_day_of_month	stays_in_weekend_nights	\
count	97261.000000	97261	
unique	NaN	18	
top	NaN	0	
freq	NaN	42629	
mean	15.777228	NaN	
std	8.792719	NaN	
min	0.000000	NaN	
25%	8.000000	NaN	
50%	16.000000	NaN	
75%	23.000000	NaN	
max	31.000000	NaN	

Columns:

```
Index(['hotel', 'is_canceled', 'lead_time', 'arrival_date_year',  
      'arrival_date_month', 'arrival_date_week_number',  
      'arrival_date_day_of_month', 'stays_in_weekend_nights',  
      'stays_in_week_nights', 'adults', 'children', 'babies', 'meal',  
      'country', 'market_segment', 'distribution_channel',  
      'is_repeated_guest', 'previous_cancellations',  
      'previous_bookings_not_canceled', 'reserved_room_type',  
      'assigned_room_type', 'booking_changes', 'deposit_type', 'agent',  
      'company', 'days_in_waiting_list', 'customer_type', 'adr',  
      'required_car_parking_spaces', 'total_of_special_requests',  
      'reservation_status', 'reservation_status_date'],  
      dtype='object')
```

===== MISSING VALUES =====

hotel	0
is_canceled	0
lead_time	0
arrival_date_year	0
arrival_date_month	0
arrival_date_week_number	0
arrival_date_day_of_month	0
stays_in_weekend_nights	0
stays_in_week_nights	0
adults	1
children	1
babies	1
meal	1
country	472
market_segment	1
distribution_channel	1
is_repeated_guest	2
previous_cancellations	2
previous_bookings_not_canceled	2
reserved_room_type	2
assigned_room_type	2
booking_changes	2
deposit_type	2
agent	13520
company	91473
days_in_waiting_list	2
customer_type	2
adr	3

Result: